

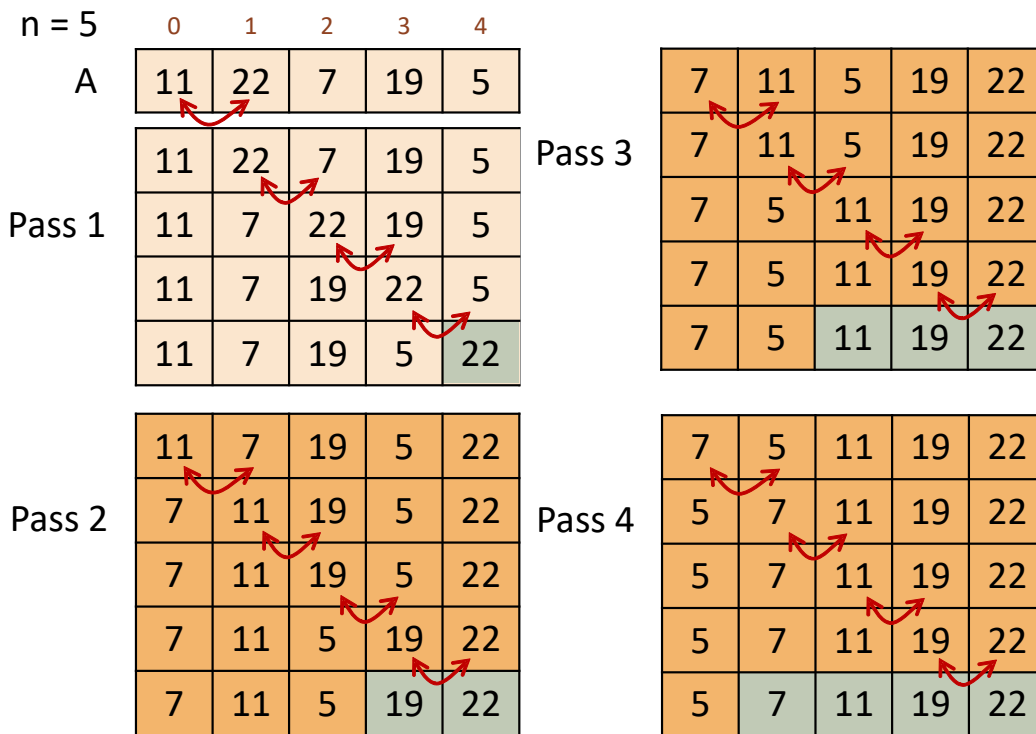
The background features a dark, textured surface with several bright, glowing lines in red and yellow. These lines are arranged to form a stylized, symmetrical figure that resembles a person with arms and legs spread wide, possibly representing a person in a dynamic pose or a stylized letter 'A'. The lines are slightly blurred, giving a sense of motion or light trails. The overall color palette is dominated by the dark background, the vibrant reds, and the warm yellows.

# Computational Thinking

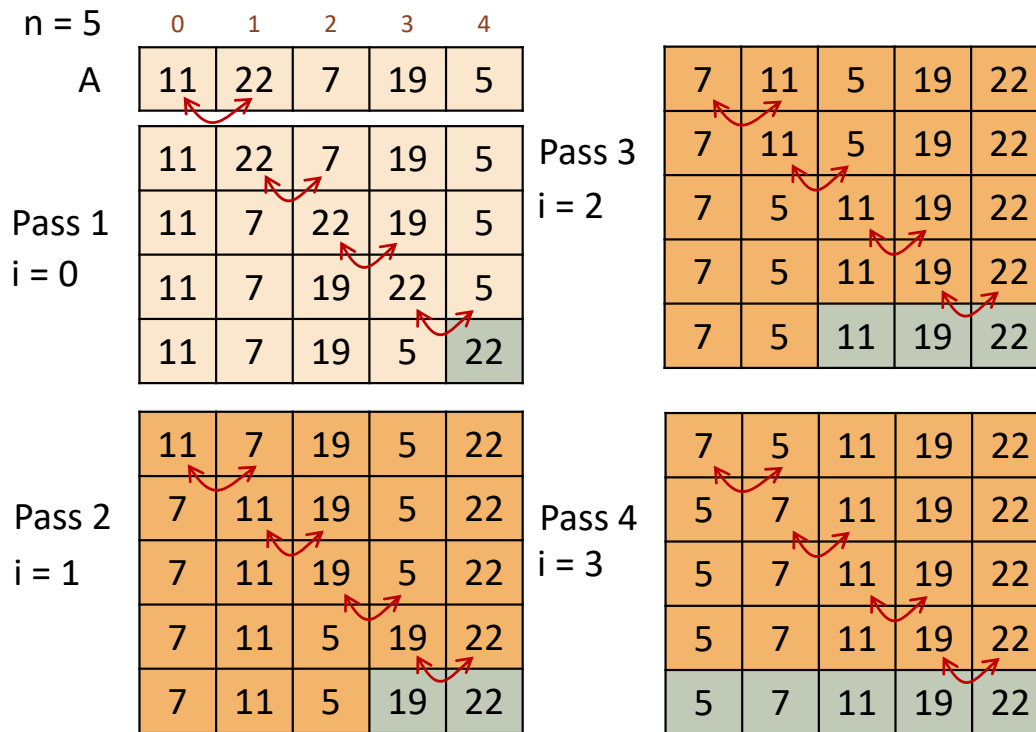
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A. BHULAI

# Bubble sort



# Bubble sort



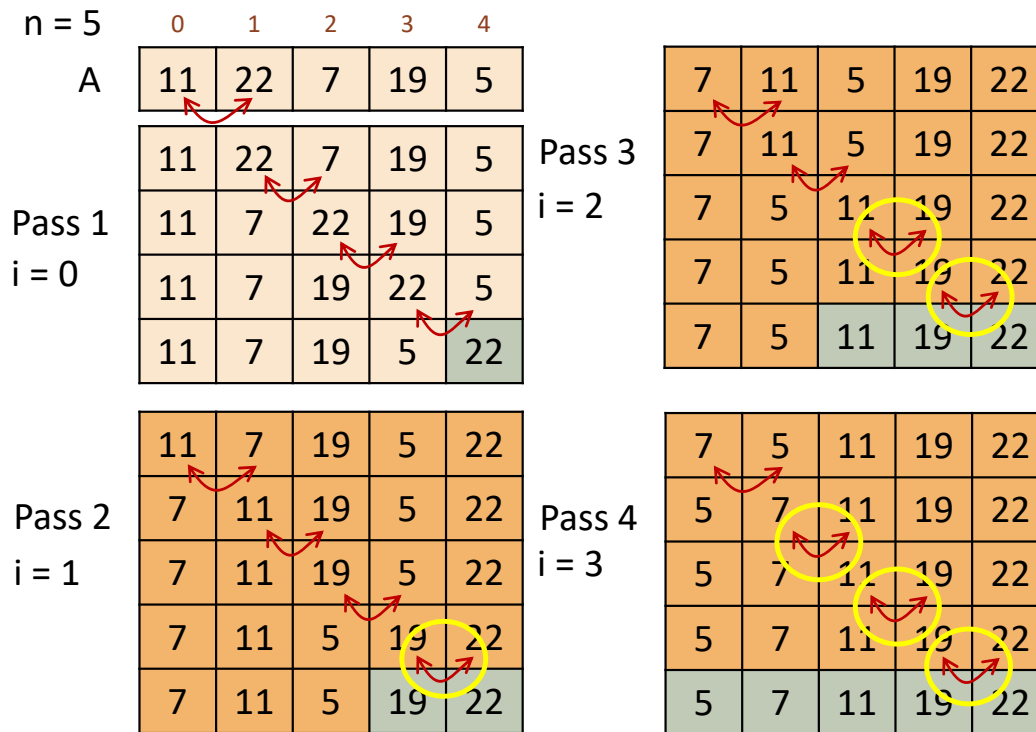
## Code

```

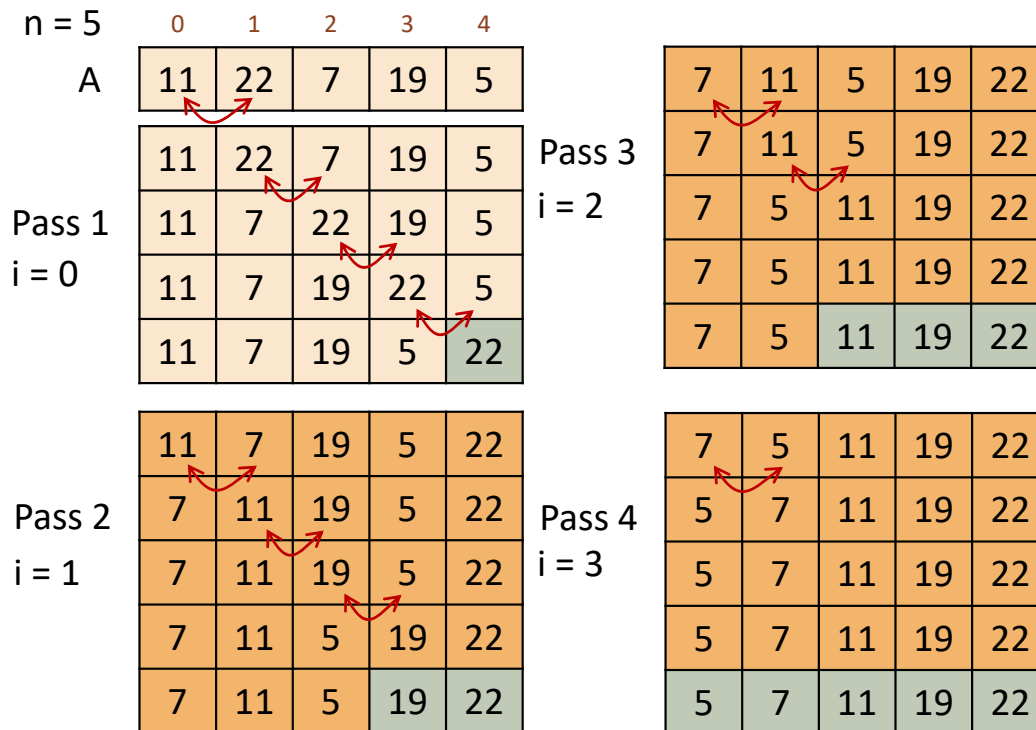
1 # Bubble sort, general case
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # Comparisons in a pass
9         for j in range(0, n-1):
10            # Swap if the element found is greater
11            # than the next element
12            if A[j] > A[j + 1]:
13                A[j], A[j + 1] = A[j + 1], A[j]
14

```

# Bubble sort



# Bubble sort



## Code

```

1 # Bubble sort
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # Last i elements are already in place
9         for j in range(0, n-i-1):
10            # Swap if the element found is greater
11            # than the next element
12            if A[j] > A[j + 1]:
13                A[j], A[j + 1] = A[j + 1], A[j]
14

```

# Bubble sort

n = 5

A

|                 | 0  | 1  | 2  | 3  | 4  |
|-----------------|----|----|----|----|----|
| Pass 1<br>i = 0 | 11 | 22 | 7  | 19 | 5  |
|                 | 11 | 22 | 7  | 19 | 5  |
|                 | 11 | 7  | 22 | 19 | 5  |
|                 | 11 | 7  | 19 | 22 | 5  |
|                 | 11 | 7  | 19 | 5  | 22 |

Pass 2  
i = 1

|  |    |    |    |    |    |
|--|----|----|----|----|----|
|  | 11 | 7  | 19 | 5  | 22 |
|  | 7  | 11 | 19 | 5  | 22 |
|  | 7  | 11 | 19 | 5  | 22 |
|  | 7  | 11 | 19 | 5  | 22 |
|  | 7  | 11 | 5  | 19 | 22 |
|  | 7  | 11 | 5  | 19 | 22 |

Pass 3  
i = 2

|  |   |    |    |    |    |
|--|---|----|----|----|----|
|  | 7 | 11 | 5  | 19 | 22 |
|  | 7 | 11 | 5  | 19 | 22 |
|  | 7 | 5  | 11 | 19 | 22 |
|  | 7 | 5  | 11 | 19 | 22 |
|  | 7 | 5  | 11 | 19 | 22 |
|  | 7 | 5  | 11 | 19 | 22 |

Pass 4  
i = 3

|  |   |   |    |    |    |
|--|---|---|----|----|----|
|  | 7 | 5 | 11 | 19 | 22 |
|  | 5 | 7 | 11 | 19 | 22 |
|  | 5 | 7 | 11 | 19 | 22 |
|  | 5 | 7 | 11 | 19 | 22 |
|  | 5 | 7 | 11 | 19 | 22 |
|  | 5 | 7 | 11 | 19 | 22 |

## Code

```

1 # Bubble sort
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # Last i elements are already in place
9         for j in range(0, n-i-1):
10            # Swap if the element found is greater
11            # than the next element
12            if A[j] > A[j + 1]:
13                A[j], A[j + 1] = A[j + 1], A[j]
14

```

Example:

Outer loop: i=2, n=5; n-1=4

Inner loop: j=0, i=2, n=5; n-i-1=2

j=0

j<2 ?? Yes

A[0] > A[1] ? No, do nothing

j=1

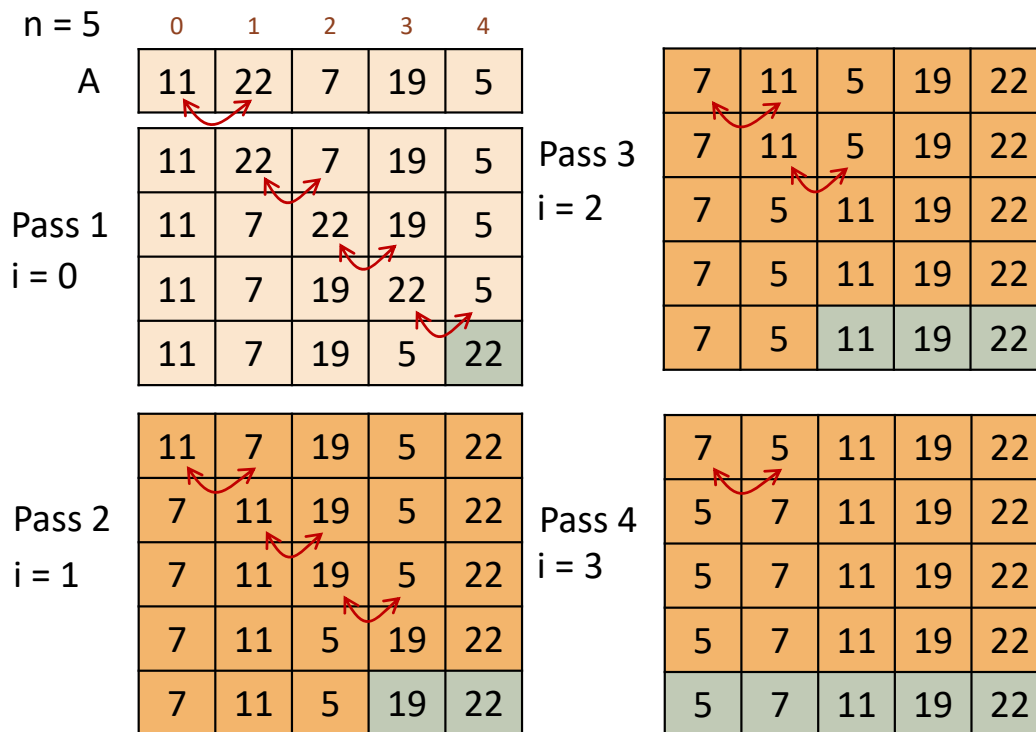
j<2 ?? Yes

A[1] > A[2] ? Yes, swap

j=2

j<2 ?? No

# Bubble sort



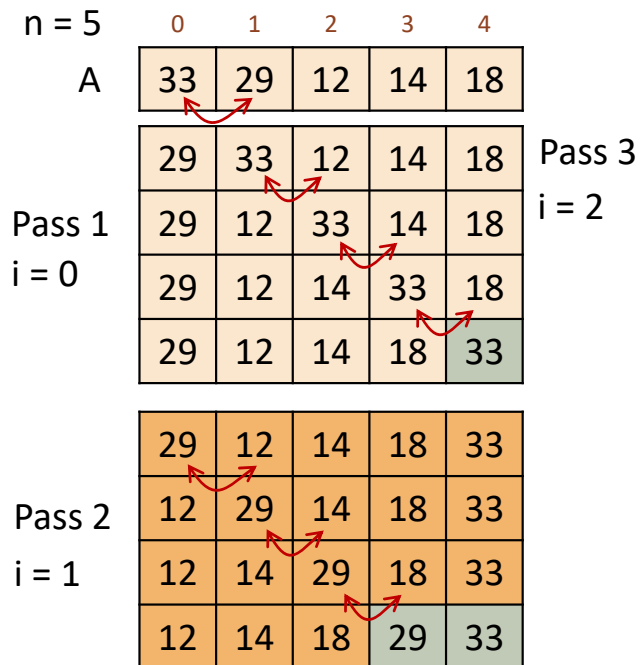
## Code

```

1 # Bubble sort
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # Last i elements are already in place
9         for j in range(0, n-i-1):
10            # Swap if the element found is greater
11            # than the next element
12            if A[j] > A[j + 1]:
13                A[j], A[j + 1] = A[j + 1], A[j]
14

```

# Bubble sort



|    |    |    |    |    |
|----|----|----|----|----|
| 12 | 14 | 18 | 29 | 33 |
| 12 | 14 | 18 | 29 | 33 |
| 12 | 14 | 18 | 29 | 33 |

no swapping

## Code

```

1 # Bubble sort
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # Last i elements are already in place
9         for j in range(0, n-i-1):
10            # Swap if the element found is greater
11            # than the next element
12            if A[j] > A[j + 1]:
13                A[j], A[j + 1] = A[j + 1], A[j]
14
15            swapped = True
16
17 if swapped==False:
18     # exit the loop if we don't need to make a swap
19     return
20
21
22

```



# Bubble sort

Example:

Outer loop:  $i=2$ ,  $n=5$ ;  $n-1=4$ ; **swapped = False**

Inner loop:  $j=0$ ,  $i=2$ ,  $n=5$ ;  $n-i-1=2$

$j=0$

$j < 2$  ?? Yes

$A[0] > A[1]$  ? No, do nothing

$j=1$

$j < 2$  ?? Yes

$A[1] > A[2]$  ? No, do nothing

$j=2$

$j < 2$  ?? No

**check swapped condition**

**swapped=False**

**exit outer loop**

Pass 3

$i = 2$

|    |    |    |    |    |
|----|----|----|----|----|
| 12 | 14 | 18 | 29 | 33 |
| 12 | 14 | 18 | 29 | 33 |
| 12 | 14 | 18 | 29 | 33 |

no swapping

## Code

```
1 # Optimized Bubble sort
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # if the array is already sorted, it doesn't need
9         # to go through the entire process
10        swapped = False
11
12        # Last i elements are already in place
13        for j in range(0, n-i-1):
14            # Swap if the element found is greater
15            # than the next element
16            if A[j] > A[j + 1]:
17                A[j], A[j + 1] = A[j + 1], A[j]
18                swapped = True
19
20        if swapped==False:
21            # exit the loop if we don't need to make a swap
22            return
```

# Bubble sort

---

## Code

```
1 # Optimized Bubble sort
2
3 def bubbleSort(A):
4     n = len(A)
5
6     # Go through all the elements in the array
7     for i in range(n-1):
8         # if the array is already sorted, it doesn't need
9         # to go through the entire process
10        swapped = False
11
12        # Last i elements are already in place
13        for j in range(0, n-i-1):
14            # Swap if the element found is greater
15            # than the next element
16            if A[j] > A[j + 1]:
17                A[j], A[j + 1] = A[j + 1], A[j]
18                swapped = True
19
20        if swapped==False:
21            # exit the loop if we don't need to make a swap
22            return
23
24 # Test code
25 A = [11, 22, 7, 19, 5]
26
27 # call the bubbleSort function
28 bubbleSort(A)
29
30 print("Sorted array is:")
31 for i in range(len(A)):
32     print("% d" % A[i], end=" ")
33
```

**THANK YOU FOR YOUR ATTENTION,  
YOU CAN CLAP NOW**



**IF YOU HAVE ANY QUESTIONS, PLEASE  
ASK MY FRIEND GOOGLE**

[makeameme.org](http://makeameme.org)